Unit 3 Guided Notes — Equations & Inequalities

ANSWER KEY

Keep it balanced. Undo step by step. Flip the sign when you divide by a negative.

🎯 **By the end of this unit, I can…**

- Solve multi-step equations, including ones with variables on both sides
- Distribute and combine like terms accurately
- Identify equations with one solution, no solution, or infinitely many

Standards: NC.8.EE.7

Answer key: every blank and final answer is filled in green.

1 One-Step Equations / Inverse Operations

📖 **Words to know**

Variable: a letter, like x, that stands for a number we do not know yet.

Coefficient: the number in front of a variable. In 3x, the coefficient is 3.

Constant: a number by itself, with no variable. In 3x + 5, the constant is 5.

Like terms: terms with the same variable part, like 3x and 2x, or 5 and 8.

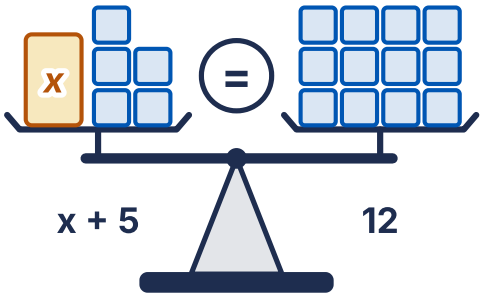
Isolate: get the variable alone on one side, like x = 4.

Solution: the number that makes the equation true.

Inequality: a math sentence with <, >, ≤ or ≥ instead of =.

📌 **THE BIG IDEA:** Use the opposite operation (the INVERSE OPERATION) to get x alone. Do the same thing to both sides. Example: x + 5 = 12 → subtract 5 from both sides → x = 7.

x tile +1 tile red = negative (−x or −1) a big block = that number



x + 5 = 12 is balanced. Keep it balanced!

Operation	Inverse	Example
Addition	Subtraction	x + 5 = 12
Subtraction	Addition	x − 3 = 8
Multiplication	Division	3x = 15
Division	Multiplication	$\frac{x}{4} = 2$

Match the Operation to its Inverse

MATCH Write the letter of each operation's inverse on the line.

B 1. Addition

D 2. Subtraction

A 3. Multiplication

C 4. Division

A. Division

B. Subtraction

C. Multiplication

D. Addition

✏ Practice: One-Step Equations

1. ★ Try it Solve: x + 9 = 15

Answer: 6

How to do it: The inverse of + 9 is − 9.
x + 9 − 9 = 15 − 9
x = 6

2. ★★ On level Solve: $\frac{x}{3} = -4$

Answer: −12

How to do it: The inverse of ÷ 3 is × 3.
$\frac{x}{3} \times 3 = -4 \times 3$
$x = -12$

3. ★★★ Challenge Mia solved $x - 8 =$ **?** and got $x = 3$. What number is the **?**

Answer: −5

How to do it: Put $x = 3$ back into the left side.
$3 - 8 = -5$
So the equation was $x - 8 = -5$.

2 Two-Step Equations

📌 **TWO STEPS:** These are the last 2 of the 5 steps. Step 1 — Undo + or −. Step 2 — Undo × or ÷.

Example 1: $-2x + 4 = 8$

SOLVE $-2x + 4 = 8$

Step 1: Get rid of the + 4. The opposite of + 4 is − 4, so subtract 4 from BOTH sides.

Step 2: Get x by itself. x is MULTIPLIED by −2. The opposite is dividing, so divide BOTH sides by −2.

Answer: $x = -2$

Example 2: $\frac{x}{2} + 6 = 7$

SOLVE $\frac{x}{2} + 6 = 7$

Step 1: Get rid of the + 6. The opposite of + 6 is − 6, so subtract 6 from BOTH sides.

Step 2: Get x by itself. x is DIVIDED by 2. The opposite is multiplying, so multiply BOTH sides by 2.

Answer: x = 2

✏ Practice: Two-Step Equations

1. ★ Try it Solve: 3x + 5 = 20

Answer: 5

How to do it: Subtract 5: 3x = 15
Divide by 3: x = 5

2. ★★ On level Solve: −3x + 4 = 19

Answer: −5

How to do it: Subtract 4: −3x = 15
Divide by −3: x = −5

3. ★★★ Challenge Leo solved 4x − 6 = 18 and wrote 4x = 12, so x = 3. Find his mistake. What is the correct answer?

Answer: 6

How to do it: Leo subtracted 6. To undo − 6 you must ADD 6.
4x − 6 + 6 = 18 + 6 → 4x = 24
Divide by 4: x = 6

3 Distributive Property

📌 **THE RULE:** Multiply the outside number by every term inside. Watch the signs: −2(x − 3) = −2x + 6.

Example: 2(2x + 3) = −6

SOLVE **2(2x + 3) = −6**

Step 1: Distribute. Multiply 2 by EACH term inside the ( ).

Step 2: Get rid of the + 6. The opposite of + 6 is − 6, so subtract 6 from BOTH sides.

Step 3: Get x by itself. x is MULTIPLIED by 4. The opposite is dividing, so divide BOTH sides by 4.

Answer: x = −3

ERROR ANALYSIS **A student expands −3(x − 4) as −3x − 12. What went wrong?**

A negative times a negative is positive: −3 · (−4) = +12. Correct: −3x + 12. Distribute the sign to EVERY term inside the parentheses.

✏ Practice: Distributive Property

1. ★ Try it Solve: 3(x + 2) = 21

Answer: 5

How to do it: Distribute: 3x + 6 = 21
Subtract 6: 3x = 15
Divide by 3: x = 5

2. ★★ On level Solve: $-2(x - 5) = 16$

Answer: **−3**

> **How to do it:** Distribute: $-2x + 10 = 16$ ($-2 \cdot -5 = +10$)
> Subtract 10: $-2x = 6$
> Divide by −2: $x = -3$

3. ★★★ Challenge Solve: $5(x - 1) - 2x = 13$

Answer: **6**

> **How to do it:** Distribute: $5x - 5 - 2x = 13$
> Combine like terms: $3x - 5 = 13$
> Add 5: $3x = 18$
> Divide by 3: $x = 6$

4 Variables on Both Sides

📌 **STRATEGY:** Move variables from the RIGHT to the LEFT using inverse operations, then solve like a two-step equation.

Example: 5x + 3 = 2x + 12

SOLVE **5x + 3 = 2x + 12**

Step 1: Move the x's to the left side. There is + 2x on the right side. Do the opposite: subtract 2x from BOTH sides. Now the x's are only on the left.

Step 2: Get rid of the + 3. The opposite of + 3 is − 3, so subtract 3 from BOTH sides.

Step 3: Get x by itself. x is MULTIPLIED by 3. The opposite is dividing, so divide BOTH sides by 3.

Answer: x = 3

✏ Practice: Variables on Both Sides

1. ★ Try it Solve: 4x + 1 = x + 10

Answer: 3

How to do it: Subtract x: 3x + 1 = 10
Subtract 1: 3x = 9
Divide by 3: x = 3

2. ★★ On level Solve: 2x + 9 = 5x − 6

Answer: 5

How to do it: Subtract 5x: −3x + 9 = −6
Subtract 9: −3x = −15
Divide by −3: x = 5

3. ★★★ Challenge Plan A costs $20 plus $3 per movie. Plan B costs $5 per movie. After how many movies do they cost the same?

Answer: 10

How to do it: Let x = number of movies.
20 + 3x = 5x
Subtract 5x: 20 − 2x = 0 → −2x = −20
Divide by −2: x = 10 movies

5 Multi-Step Equations — The 5-Step Process

📌 **THE 5 STEPS:** 1. Distribute → 2. Combine Like Terms → 3. Move Variables → 4. Inverse for + or − → 5. Inverse for × or ÷

Example: $-2x + 1 = 8(x + 2) - 5x$

SOLVE **$-2x + 1 = 8(x + 2) - 5x$**

Step 1: Distribute. Multiply 8 by EACH term inside the ( ).

Step 2: Combine like terms. Put like terms together (x terms with x terms, numbers with numbers): 8x − 5x = 3x.

Step 3: Move the x's to the left side. There is + 3x on the right side. Do the opposite: subtract 3x from BOTH sides. Now the x's are only on the left.

Step 4: Get rid of the + 1. The opposite of + 1 is − 1, so subtract 1 from BOTH sides.

Step 5: Get x by itself. x is MULTIPLIED by −5. The opposite is dividing, so divide BOTH sides by −5.

Answer: x = −3

✏ Practice: Multi-Step Equations

1. ★ Try it Solve: 2(x + 3) + 4x = 30

Answer: 4

How to do it: Distribute: 2x + 6 + 4x = 30
Combine: 6x + 6 = 30
Subtract 6: 6x = 24
Divide by 6: x = 4

2. ★★ On level Solve: 3(x − 2) = x + 8

Answer: 7

How to do it: Distribute: 3x − 6 = x + 8
Subtract x: 2x − 6 = 8
Add 6: 2x = 14
Divide by 2: x = 7

3. ★★★ Challenge Solve: 4(x + 1) − 2x = 3(x − 2) + 7

Answer: 3

How to do it: Distribute: 4x + 4 − 2x = 3x − 6 + 7
Combine: 2x + 4 = 3x + 1
Subtract 3x: −x + 4 = 1
Subtract 4: −x = −3
Divide by −1: x = 3

6 Special Cases — No Solution & Infinite Solutions

🔑 **NO SOLUTION:** Variables cancel and you get a FALSE statement (like 5 = 9).

🔑 **INFINITE SOLUTIONS:** Variables cancel and you get a TRUE statement (like 6 = 6).

Result	Meaning	Example
Variables stay; x = number	One Solution	2x+1=7 → x=3
Variables cancel; FALSE remains	No Solution	2x+1=2x+5 → 1=5 ×
Variables cancel; TRUE remains	Infinite Solutions	2x+1=2x+1 → 1=1 ✓

No Solution — Solve: $3(x + 2) = 3x + 10$

SOLVE $3(x + 2) = 3x + 10$

Step 1: Distribute. Multiply 3 by EACH term inside the ( ).

Step 2: Move the x's to the left side. There is + 3x on the right side. Do the opposite: subtract 3x from BOTH sides. Now the x's are only on the left.

Answer: No solution

Infinite Solutions — Solve: $2(x + 4) = 2x + 8$

SOLVE $2(x + 4) = 2x + 8$

Step 1: Distribute. Multiply 2 by EACH term inside the ( ).

Step 2: Move the x's to the left side. There is + 2x on the right side. Do the opposite: subtract 2x from BOTH sides. Now the x's are only on the left.

Answer: Infinitely many solutions

Quick Check

MC **Solving an equation gives 7 = 7. The equation has...**

- ○ One solution
- ○ No solution
- ● **Infinitely many solutions**

✏ Practice: Special Cases

1. ★ Try it You solve an equation. The x's cancel and you get 2 = 9. How many solutions?

- ○ One solution
- ● **No solution**
- ○ Infinitely many solutions

> **How to do it:** 2 = 9 is FALSE. A false statement means no number works → No solution.

2. ★★ On level How many solutions does 5(x − 2) = 5x − 10 have?

- ○ One solution
- ○ No solution
- ● **Infinitely many solutions**

> **How to do it:** Distribute: 5x − 10 = 5x − 10
> Subtract 5x: −10 = −10. TRUE → Infinitely many solutions.

3. ★★ On level How many solutions does 3x + 1 = 2x + 1 have?

- ● **One solution**
- ○ No solution
- ○ Infinitely many solutions

> **How to do it:** Subtract 2x: x + 1 = 1
> Subtract 1: x = 0. The x's did NOT cancel → One solution (x = 0).

4. ★★★ Challenge What number goes in the box so 2(x + 5) = 2x + ☐ has **infinitely many** solutions?

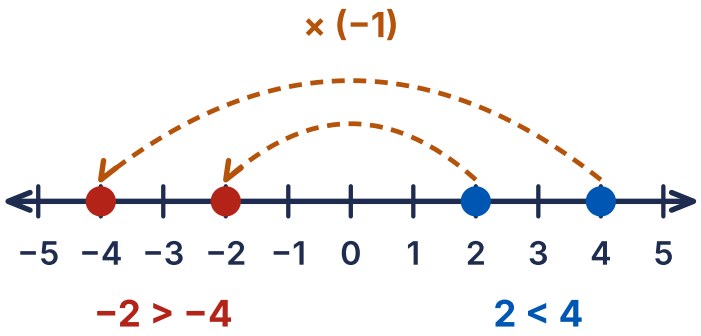
Answer: 10

> **How to do it:** Distribute: 2(x + 5) = 2x + 10.
> For a TRUE statement, both sides must be the same: 2x + 10 = 2x + 10. The box is 10.

7 Linear Inequalities

Symbol	Meaning	Circle	Shade
<	less than	OPEN	LEFT
>	greater than	OPEN	RIGHT
≤	less than or equal	CLOSED	LEFT
≥	greater than or equal	CLOSED	RIGHT

⚠️ **CRITICAL RULE — FLIP THE SIGN:** When you MULTIPLY or DIVIDE both sides by a NEGATIVE number, you must FLIP the inequality sign! Example: −2x > 6 → divide by −2 → x < −3 (sign flipped!)



The order flips, so the sign flips.

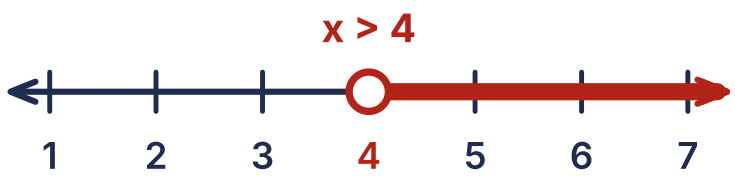
Example (no flip): $2x + 3 > 11$

SOLVE $2x + 3 > 11$

Step 1: Get rid of the + 3. The opposite of + 3 is − 3, so subtract 3 from BOTH sides.

Step 2: Get x by itself. x is MULTIPLIED by 2. The opposite is dividing, so divide BOTH sides by 2.

Answer: $x > 4$



x > 4: open circle (4 is not included)

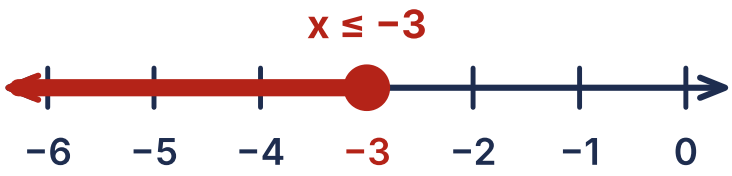
Example (FLIP NEEDED): $-3x + 4 \geq 13$

SOLVE $-3x + 4 \geq 13$

Step 1: Get rid of the + 4. The opposite of + 4 is − 4, so subtract 4 from BOTH sides.

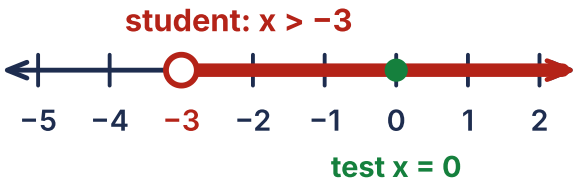
Step 2: Get x by itself. x is MULTIPLIED by −3. The opposite is dividing, so divide BOTH sides by −3. −3 is negative, so the inequality sign FLIPS.

Answer: $x \leq -3$



x ≤ −3: closed circle (sign flipped!)

ERROR ANALYSIS To solve −4x > 12 a student wrote **x > −3**. What's the mistake?



Test x = 0 in −4x > 12. Is −4(0) > 12 true?

Dividing both sides by the negative number −4 requires FLIPPING the sign. Correct: x < −3.

✏ Practice: Linear Inequalities

1. ★ Try it Solve: x + 6 ≤ 10.

Answer: x ≤ 4

How to do it: Subtract 6: x ≤ 4. (No flip needed.)

2. ★ Try it What inequality does this number line show?



Answer: x ≥ 2

How to do it: Closed circle at 2 → 2 is included. Shaded right → greater. x ≥ 2

3. ★★ On level Solve: −2x + 5 > 11.

Answer: x < −3

> **How to do it:** Subtract 5: −2x > 6
> Divide by −2 and FLIP: x < −3

4. ★★★ Challenge Dev solved −5x ≤ 20 and got x ≤ −4. Find his mistake. What is the correct answer?

Answer: x ≥ −4

> **How to do it:** Dev divided by −5 but forgot to FLIP the sign.
> −5x ≤ 20 → divide by −5 and flip → x ≥ −4

8 Word Problems — Equations & Inequalities

📌 **WORD PROBLEMS IN 4 STEPS:** 1) Let x = the thing you don't know. → 2) Write the equation. → 3) Solve it with the 5 steps. → 4) Answer in a sentence.

Equation (=)	Inequality	Addition	Subtraction
total, equals, is, the same as, gives, results in	at least (≥), at most (≤), more than (>), fewer than (<)	sum, plus, more than, increased by, total	difference, minus, less than, decreased by

SHORT ANSWER Movie Theater: A movie theater charges a $6 entry fee plus $4 per ticket. If the total cost was $30, how many tickets were purchased?

Let x = number of tickets

Equation: 6 + 4x = 30

Solve (show each step):

Answer: 6 tickets were purchased.

SHORT ANSWER Notebooks: A student wants to spend no more than $50 on notebooks that cost $6 each. How many notebooks can the student buy?

Let x = number of notebooks

Equation: 6x ≤ 50

Solve (show each step):

Answer: x must be a whole number, so the student can buy **at most 8 notebooks.**

✏ Practice: Word Problems

1. ★ Try it Tickets cost $7 each. Maya spent $42. How many tickets did she buy?

Answer: 6

> **How to do it:** Let x = number of tickets.
> 7x = 42
> Divide by 7: x = 6 tickets

2. ★★ On level Which equation matches: "Twice a number, decreased by 5, is 17"?

- ● **2x − 5 = 17**
- ○ 2(x − 5) = 17
- ○ 5 − 2x = 17
- ○ 2x + 5 = 17

> **How to do it:** Twice a number → 2x. Decreased by 5 → 2x − 5. Is 17 → = 17. So 2x − 5 = 17 (and x = 11).

3. ★★ On level A plumber charges $50 plus $30 per hour. The bill was $170. How many hours did the plumber work?

Answer: 4

> **How to do it:** Let x = hours.
> 50 + 30x = 170
> Subtract 50: 30x = 120
> Divide by 30: x = 4 hours

4. ★★★ Challenge Jordan has $25. There is a $3 fee, and each snack costs $4. What is the greatest number of snacks Jordan can buy?

Answer: 5

> **How to do it:** Let x = snacks. "Has $25" means the cost is at most 25.
> 3 + 4x ≤ 25
> Subtract 3: 4x ≤ 22
> Divide by 4: x ≤ 5.5
> You can't buy half a snack, so the greatest number is 5.

✏ Part 2 — Comprehensive Practice

📌 **Show every step on its own line.** Box your final answer.

This part mixes problems from **all 8 sections**. Every problem has a level: ★ Try it ★★ On level ★★★ Challenge. Stuck? Do the ★ problems first, then look back at that section of the notes.

A. Multi-Step Equations Sections 2–5

1) ★ Try it $3x + 7 = 22$

SOLVE $3x + 7 = 22$

Answer: x = 5

2) ★ Try it $2(x - 4) = 10$

SOLVE $2(x - 4) = 10$

Answer: x = 9

3) ★★ On level $5x - 3 = 2x + 9$

SOLVE $5x - 3 = 2x + 9$

Answer: x = 4

4) ★★ On level $-4(x + 2) = 12$

SOLVE $-4(x + 2) = 12$

Answer: x = −5

5) ★★ On level $6x + 4 = 4x - 8$

SOLVE $6x + 4 = 4x - 8$

Answer: $x = -6$

6) ★★★ Challenge $3(2x + 1) = 4x + 7$

SOLVE $3(2x + 1) = 4x + 7$

Answer: $x = 2$

7) ★ Try it $\frac{x}{4} + 5 = 8$

SOLVE $\frac{x}{4} + 5 = 8$

Answer: $x = 12$

8) ★★ On level $7 - 2x = 3x + 17$

SOLVE $7 - 2x = 3x + 17$

Answer: $x = -2$

9) ★★ On level $0.5x + 3 = 7$

SOLVE $0.5x + 3 = 7$

Answer: x = 8

10) ★★★ Challenge $-(x - 6) = 2x + 3$

SOLVE $-(x - 6) = 2x + 3$

Answer: x = 1

B. Special Cases — No Solution or Infinite Solutions? Section 6

1) ★ Try it $2x + 3 = 2x + 3$

SOLVE $2x + 3 = 2x + 3$

Answer:
Infinitely many solutions

2) ★ Try it $4x + 1 = 4x - 5$

SOLVE $4x + 1 = 4x - 5$

Answer:
No solution

3) ★★ On level $3(x + 2) = 3x + 6$

SOLVE $3(x + 2) = 3x + 6$

Answer: Infinitely many solutions

4) ★★ On level $5x - 5 = 5(x + 1)$

SOLVE $5x - 5 = 5(x + 1)$

Answer: No solution

5) ★ Try it $7x + 2 = 7x + 2$

SOLVE $7x + 2 = 7x + 2$

Answer: Infinitely many solutions

6) ★ Try it $x + 4 = x - 1$

SOLVE $x + 4 = x - 1$

Answer: No solution

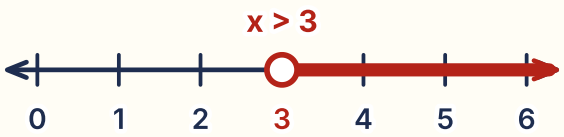
C. Inequalities Section 7

Solve it like an equation. FLIP the sign only when you multiply or divide both sides by a negative number.

1) ★ Try it $3x + 2 > 11$

SOLVE $3x + 2 > 11$

Answer: $x > 3$

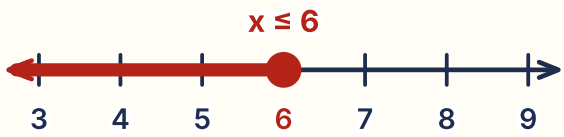


x > 3: open circle at 3, shade right

2) ★ Try it $2x - 5 \leq 7$

SOLVE $2x - 5 \leq 7$

Answer: $x \leq 6$

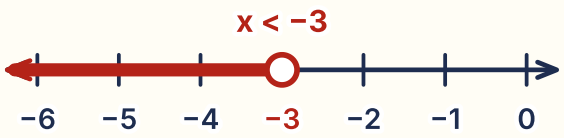


x ≤ 6: closed circle at 6, shade left

3) ★★ On level $-4x > 12$ (flip!)

SOLVE $-4x > 12$

Answer: $x < -3$

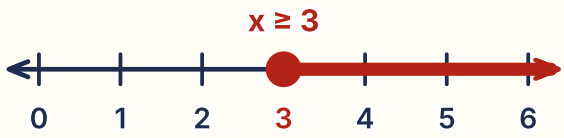


x < −3: open circle at −3, shade left

4) ★ Try it $5x + 1 \geq 16$

SOLVE $5x + 1 \geq 16$

Answer: $x \geq 3$

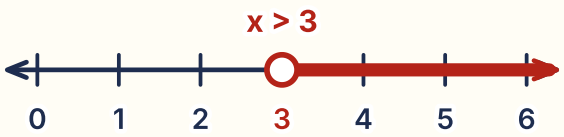


x ≥ 3: closed circle at 3, shade right

5) ★★★ Challenge 7 − x < 4 (flip!)

SOLVE $7 - x < 4$

Answer: x > 3

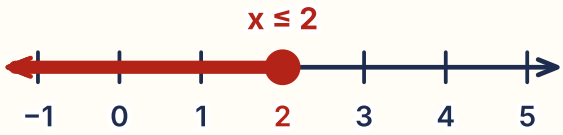


x > 3: open circle at 3, shade right

6) ★★ On level 2(x + 3) ≤ 10

SOLVE $2(x + 3) \leq 10$

Answer: x ≤ 2

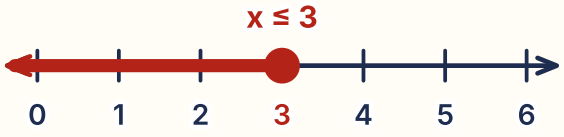


x ≤ 2: closed circle at 2, shade left

7) ★★★ Challenge −3x + 9 ≥ 0 (flip!)

SOLVE $-3x + 9 \geq 0$

Answer: x ≤ 3

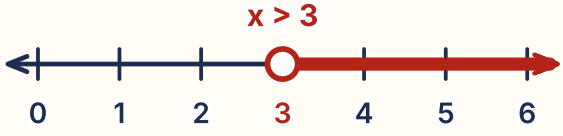


x ≤ 3: closed circle at 3, shade left

8) ★★ On level 4x − 1 > 2x + 5

SOLVE $4x - 1 > 2x + 5$

Answer: x > 3



x > 3: open circle at 3, shade right

D. Word Problems Section 8

1. ★★ On level A phone plan costs $35 per month plus $0.10 per text. Last month's bill was $52. How many texts were sent?

Let x = number of texts

Equation: 35 + 0.10x = 52

Solve (show each step):

Answer: 170 texts were sent.

2. ★★★ Challenge Sam has $240 and saves $30 per week. Jake has $420 and spends $10 per week. When will they have the same amount?

Let x = number of weeks

Equation: 240 + 30x = 420 − 10x

Solve (show each step):

Answer: They will have the same amount after 4.5 weeks.

3. ★★★ Challenge A rectangle's perimeter is 56 cm. The length is 4 cm more than the width. Find the width.

Let x = width (cm), so the length is x + 4

Equation: 2(x + 4) + 2x = 56

Solve (show each step):

Answer: The width is 12 cm.

4. ★★ On level Movie tickets cost $9 each. You have at most $50 to spend. What is the maximum number of tickets?

Let x = number of tickets

Equation: 9x ≤ 50

Solve (show each step):

Answer: x must be a whole number, so you can buy **at most 5 tickets.**

5. ★★★ Challenge A car rental costs $45 per day plus $0.25 per mile. Your budget is $120 for one day. How many miles can you drive?

Let x = number of miles

Equation: 45 + 0.25x ≤ 120

Solve (show each step):

Answer: You can drive **at most 300 miles.**

6. ★★★ Challenge Ana, Ben and Cal earned $78 in all. Ben earned $2 more than Ana. Cal earned $4 more than Ana. How much did Ana earn?

Let x = Ana's money, so Ben has x + 2 and Cal has x + 4

Equation: x + (x + 2) + (x + 4) = 78

Solve (show each step):

Answer: Ana earned **$24** (Ben earned $26 and Cal earned $28).

E. Mixed Review Sections 1–8

✏ Comprehensive Practice

1. ★ Try it Section 1 · Solve: x − 7 = −2

Answer: 5

> **How to do it:** Add 7: x = −2 + 7 = 5

2. ★ Try it Section 2 · Solve: 5x − 4 = 26

Answer: 6

> **How to do it:** Add 4: 5x = 30
> Divide by 5: x = 6

3. ★ Try it Section 7 · Which inequality means "x is at most 12"?

- ● **x ≤ 12**
- ○ x < 12
- ○ x ≥ 12
- ○ x > 12

> **How to do it:** At most 12 → 12 is allowed, and anything less → x ≤ 12.

4. ★★ On level Section 3 · Solve: −3(x + 4) = 6

Answer: −6

> **How to do it:** Distribute: −3x − 12 = 6
> Add 12: −3x = 18
> Divide by −3: x = −6

5. ★★ On level Section 4 · Solve: 9x − 2 = 4x + 23

Answer: 5

> **How to do it:** Subtract 4x: 5x − 2 = 23
> Add 2: 5x = 25
> Divide by 5: x = 5

6. ★★ On level Section 6 · How many solutions does 3(2x − 1) = 6x + 3 have?

- ○ One solution
- ● **No solution**
- ○ Infinitely many solutions

> **How to do it:** Distribute: 6x − 3 = 6x + 3
> Subtract 6x: −3 = 3. FALSE → No solution.

7. ★★ On level Section 7 · Solve: 8 − 3x ≤ 20.

Answer: x ≥ −4

How to do it: Subtract 8: −3x ≤ 12
Divide by −3 and FLIP: x ≥ −4

8. ★★★ Challenge Section 5 · Solve: 2(3x − 1) + 4 = 4(x + 2)

Answer: 3

How to do it: Distribute: 6x − 2 + 4 = 4x + 8
Combine: 6x + 2 = 4x + 8
Subtract 4x: 2x + 2 = 8
Subtract 2: 2x = 6
Divide by 2: x = 3

9. ★★★ Challenge Section 8 · The sum of three times a number and 8 is −13. What is the number?

Answer: −7

How to do it: 3x + 8 = −13
Subtract 8: 3x = −21
Divide by 3: x = −7

10. ★★★ Challenge Section 8 · Ava has 18 stickers and gets 4 more each day. Ben has 50 stickers and gives away 4 each day. After how many days do they have the same number?

Answer: 4

How to do it: Let x = days.
18 + 4x = 50 − 4x
Add 4x: 18 + 8x = 50
Subtract 18: 8x = 32
Divide by 8: x = 4 days (both have 34 stickers)